

COUNTEREXAMPLES TO THE FORMULA

$$\mathbf{g}(C_3 \oplus C_{3p}) = 3p + 3$$

ABSTRACT. For every positive integer $n \equiv 5 \pmod{6}$, we construct a $(3n + 3)$ -element subset of $C_3 \oplus C_{3n}$ containing no zero-sum subset of cardinality $3n$. Hence

$$\mathbf{g}(C_3 \oplus C_{3n}) \geq 3n + 4.$$

For every prime $p \equiv 5 \pmod{6}$, this contradicts Theorem 3.1 of Guillot, Marchan, Ordaz, Schmid, and Zerdoum, which asserts $\mathbf{g}(C_3 \oplus C_{3p}) = 3p + 3$ for all primes $p \neq 3$. The smallest counterexample produced by this construction is $p = 5$. We also identify the erroneous step in the proof of their Proposition 3.7.

Let C_m denote the cyclic group of order m , written additively. For a finite abelian group G , its Harborth constant $\mathbf{g}(G)$ is the least integer k such that every subset of G of cardinality at least k contains a zero-sum subset of cardinality $\exp(G)$; see Bajnok [1, Chapter F.3]. Guillot et al. [2, Theorem 3.1] state that

$$\mathbf{g}(C_3 \oplus C_{3p}) = 3p + 3$$

for every prime $p \neq 3$. The following construction disproves this formula for every prime $p \equiv 5 \pmod{6}$ and applies more generally to all integers in that congruence class.

Theorem 1. *If $n \equiv 5 \pmod{6}$, then*

$$\mathbf{g}(C_3 \oplus C_{3n}) \geq 3n + 4.$$

The bound $\mathbf{g}(C_3 \oplus C_{3n}) \geq 3n + 3$ for $n \geq 2$ is due to Kiefer; see also [1, Prop. F.104]. Theorem 1 improves it by one for $n \equiv 5 \pmod{6}$.

Proof. Write $n = 3m - 1$, so $m = (n + 1)/3$ is even. Define

$$\rho : C_3 \oplus C_{3n} \longrightarrow C_n, \quad \rho(a, b) = b \pmod{n},$$

and, using the integer representatives in $\{0, \dots, n - 1\}$, set

$$J = \{m, m + 1, \dots, 2m - 1\} \subset C_n, \quad A_n = \rho^{-1}(J).$$

Every fiber of ρ has cardinality 9, and therefore

$$|A_n| = 9m = 3n + 3.$$

For $j \in J$ one has

$$\rho^{-1}(j) = \{(a, j + \ell n) : a, \ell \in \{0, 1, 2\}\},$$

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where the second coordinate is taken in C_{3n} . Hence the sum of this fiber is $(0, 9j)$, and

$$\sigma(A_n) = \left(0, 9 \sum_{j=m}^{2m-1} j\right) = \left(0, \frac{9mn}{2}\right) = 0 \quad \text{in } C_3 \oplus C_{3n},$$

since m is even and $9mn/2$ is divisible by $3n$.

Suppose that $B \subset A_n$ has cardinality $3n$ and sum 0, and put $R = A_n \setminus B$. Then $|R| = 3$ and

$$\sigma(R) = \sigma(A_n) - \sigma(B) = 0.$$

For each $r \in R$, let $\widetilde{\rho(r)} \in J$ be the chosen integer representative of $\rho(r)$. Then

$$n + 1 = 3m \leq \sum_{r \in R} \widetilde{\rho(r)} \leq 3(2m - 1) = 2n - 1.$$

This interval contains no multiple of n , so $\rho(\sigma(R)) \neq 0$, a contradiction. Thus A_n contains no zero-sum subset of cardinality $3n$. Since $\exp(C_3 \oplus C_{3n}) = 3n$, the claimed lower bound follows. $\square$

Corollary 2. *For every prime $p \equiv 5 \pmod{6}$,*

$$\mathbf{g}(C_3 \oplus C_{3p}) \geq 3p + 4.$$

In particular,

$$\mathbf{g}(C_3 \oplus C_{15}) \geq 19,$$

contradicting the value 18 asserted in [2, Theorem 3.1]. Moreover, taking $n = 35$ gives

$$\mathbf{g}(C_3 \oplus C_{105}) \geq 109,$$

which disproves the suggested extension of the formula to composite parameters in [2, p. 632].

The construction establishes these lower bounds only; it does not determine the exact Harborth constants in the affected cases.

Remark 3 (Failure of Proposition 3.7). Let $p \equiv 5 \pmod{6}$ be prime and write $p = 3m - 1$. In the decomposition $C_3 \oplus C_{3p} = H_1 \oplus H_2$ used in [2], one has $H_1 = \ker \rho \cong C_3^2$, and the restriction of ρ identifies H_2 with C_p . Viewing the set A_p above as a squarefree sequence, every H_1 -fiber has H_2 -coordinate set

$$J = \{m, m + 1, \dots, 2m - 1\}.$$

Moreover, its H_1 -support is all of H_1 and

$$\sigma(\pi_1(A_p)) = m \sum_{h \in H_1} h = 0.$$

Thus A_p satisfies the hypotheses of [2, Proposition 3.7], while Theorem 1 shows that it violates the conclusion.

The error occurs near the end of the proof of that proposition. For an arithmetic progression

$$P_h = \{s_h e, (s_h + 1)e, \dots, (s_h + m - 1)e\} \subset C_p,$$

write $\Sigma_3(P_h)$ for the set of sums of three distinct terms. Its elements have integer coefficients in

$$[3s_h + 3, 3s_h + p - 5].$$

The proof infers from $0 \notin \Sigma_3(P_h)$ that the upper endpoint is less than p . This overlooks the possibility that the entire interval lies strictly between p and $2p$. For the fibers above, $s_h = m$, and for $p \geq 11$ the interval is

$$[p + 4, 2p - 4],$$

which contains no multiple of p . For $p = 5$ one has $m = 2$, so each progression P_h has only two terms and $\Sigma_3(P_h)$ is empty. Proposition 3.7 is used in the proof of [2, Theorem 3.1].

REFERENCES

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